

**** CHAPTER 3 —**

REVERSING A NUMBER**

**** SLIDE 1 —**

WHAT IS REVERSING A NUMBER?**

MEET NUMBER REVERSING!

Normally, we read a number from **left to right**.

For example:

1234

But what if we want to read the digits in the **opposite order**?

4321

Simple Definition

Reversing a number means arranging its digits in the opposite order.

LOOK AT THIS!

1234

1 → 2 → 3 → 4

After reversing:

4 → 3 → 2 → 1

So:

1234 → 4321

EASY ANALOGY

Think of a number like a line of people 

1 → 2 → 3 → 4

Now everyone turns around:

4 → 3 → 2 → 1

That's a **reversed number**.

★ **REMEMBER!**

Original → 1234

Reverse → 4321

****  SLIDE 2 — **

GETTING DIGITS FOR REVERSING**

🤔 **HOW DO WE REVERSE A NUMBER?**

To reverse a number, we need to get its digits **one by one**.

We already know:

% 10 → Get the last digit

 **GET THE LAST DIGIT**

`digit = n % 10;`

For example:

`n = 1234`

`1234 % 10 = 4`

👉 `Digit = 4`

 **LOOK AT THIS!**

1234 → 4

123 → 3

12 → 2

1 → 1

The digits are accessed as:

4 → 3 → 2 → 1

 **HOW DO WE REMOVE EACH DIGIT?**

After getting the digit:

$n = n / 10;$

Example:

$1234 / 10 = 123$

★ **REMEMBER!**

% 10 → GET 

/ 10 → REMOVE 

****  SLIDE 3 — **

BUILDING THE REVERSE**

 **HOW DO WE BUILD A NEW NUMBER?**

We need to take each digit and **add it to a new number**.

We use:

$rev = rev * 10 + digit;$

 **EXAMPLE**

Let's reverse:

1234

Start with:

rev = 0

STEP 1

Digit = 4

rev = $0 \times 10 + 4$

rev = 4

STEP 2

Digit = 3

rev = $4 \times 10 + 3$

rev = 43

STEP 3

Digit = 2

rev = $43 \times 10 + 2$

rev = 432

STEP 4

Digit = 1

rev = $432 \times 10 + 1$

rev = 4321

 **SO:**

1234 → 4321

★ REMEMBER!

$rev * 10 + digit \rightarrow$ Builds the reverse 

**  SLIDE 4 — 

REVERSING A NUMBER USING A LOOP**

🤔 CAN WE DO IT AUTOMATICALLY?

Yes! 

We can use a **while loop** to repeat the process.

 **BASIC CODE**

```
int n, digit;
int rev = 0;
while (n != 0)
{
    digit = n % 10;
    rev = rev * 10 + digit;
    n = n / 10;
}
```

 **HOW DOES IT WORK?**

For $n = 1234$:

$1234 \rightarrow digit = 4 \rightarrow rev = 4$

$123 \rightarrow digit = 3 \rightarrow rev = 43$

$12 \rightarrow digit = 2 \rightarrow rev = 432$

1 → digit = 1 → rev = 4321

BIG MEMORY TRICK

% 10 → GET 

/ 10 → REMOVE 

rev * 10 + digit → BUILD 

★ TAKEAWAY

Get the digit → Build the reverse → Remove the digit → Repeat 

** SLIDE 5 —

REVERSING A NUMBER RECAP**

WHAT DID WE LEARN?

1. GET THE LAST DIGIT

digit = n % 10;

% 10 → Gets the last digit

2. REMOVE THE LAST DIGIT

n = n / 10;

/ 10 → Removes the last digit

3. BUILD THE REVERSE

rev = rev * 10 + digit;

 Adds the digit to the reversed number.

 ONE BIG MEMORY TRICK!

% 10



GET 

/ 10



REMOVE 

rev * 10 + digit



BUILD 

Question

❏ What is the reverse of 1234?

- A) 1243
- B) 4321
- C) 4312
- D) 1234

❏ Which statement builds the reverse?

- A) rev = rev + digit;
- B) rev = rev * 10 + digit;
- C) rev = digit / 10;
- D) rev = n / 10;

❏ Which statement is used to initialize the reverse number?

- A) `int rev = 1;`
- B) `int rev = 0;`
- C) `int rev = 10;`
- D) `int rev = n;`

 Hint: Before building the reverse, we need an empty starting value.

 Glossary:

Initialize → Giving a variable its starting value.

3 Complete the code

```
int n = 4321;
```

```
int digit;
```

```
digit = n _____ 10;
```

 Hint: We want to GET the last digit.

 Glossary:

`% 10` → Gets the last digit.

4 Complete the code

```
int n = 4321;
```

```
n = n _____ 10;
```

 Hint: We want to REMOVE the last digit.

 Glossary:

`/ 10` → Removes the last digit from an integer.

5 Complete the Reverse Statement

```
int rev = 0;
```

```
int digit = 5;
```

```
rev = rev _____ 10 + digit;
```



Hint: Use the statement that builds the reverse.



Glossary:

rev → Variable used to store the reversed number.